

MCMaster UNIVERSITY, ON, Canada

WMO: 712975

Lat: 43.2619N

Lon: 79.9204W

Elev: 114

StdP: 99.96

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.3	-14.5	-23.3	0.5	-16.8	-20.4	0.6	-13.3	8.1	0.3	7.6	-0.6	2.3	260	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	31.1	21.1	29.4	20.6	28.0	19.9	23.5	27.9	22.5	26.8	21.6	25.7	3.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.2	17.1	25.5	21.2	16.0	24.7	20.2	15.1	23.9	70.6	27.7	66.8	26.8	63.6	25.7	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.7	6.9	6.1	-20.9	33.3	3.0	1.6	-23.0	34.4	-24.8	35.3	-26.5	36.2	-28.7	37.4		
(5)			-21.3	25.2	2.9	1.2	-23.4	26.0	-25.1	26.7	-26.7	27.4	-28.8	28.2	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.1	-4.3	-3.4	1.0	6.8	14.1	18.8	22.1	21.0	17.4	10.9	4.3	-0.6	(6)
(7)		DBStd	10.30	5.97	5.91	5.61	4.51	4.76	3.66	2.96	2.80	3.94	4.38	4.58	4.64	(7)
(8)		HDD10.0	1777	443	375	282	117	13	0	0	0	0	41	177	329	(8)
(9)		HDD18.3	3694	700	608	537	346	148	38	4	8	65	233	420	587	(9)
(10)		CDD10.0	1447	1	0	4	21	141	265	374	343	221	69	7	0	(10)
(11)		CDD18.3	322	0	0	0	0	18	52	119	92	36	3	0	0	(11)
(12)		CDH23.3	2347	0	0	1	9	176	401	917	577	245	22	0	0	(12)
(13)		CDH26.7	618	0	0	0	1	45	110	272	123	64	3	0	0	(13)
(14)	Wind	WSAvg	2.7	3.0	3.1	2.9	3.2	2.7	2.6	2.4	2.3	2.3	2.7	2.7	3.0	(14)
(15)	Precipitation	PrecAvg	884	71	62	64	76	80	79	82	77	77	73	73	69	(15)
(16)		PrecMax	1102	148	115	140	145	146	196	187	159	173	150	133	137	(16)
(17)		PrecMin	686	30	33	27	31	26	23	18	30	25	17	21	36	(17)
(18)		PrecStd	109	29	22	31	30	30	40	42	31	34	33	28	25	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.0	14.6	17.9	24.6	30.6	31.7	33.4	31.2	31.3	26.0	18.7	13.1	(19)
(20)			MCWB	11.6	10.2	11.7	14.8	19.7	20.9	22.4	21.7	22.5	18.2	13.2	10.5	(20)
(21)		2%	DB	9.3	10.3	14.9	20.6	27.9	29.9	31.5	29.5	28.6	22.7	16.0	10.2	(21)
(22)			MCWB	7.5	7.4	8.9	13.1	18.3	20.3	21.4	21.2	21.4	17.2	11.9	8.3	(22)
(23)		5%	DB	5.7	6.7	11.9	17.5	25.2	27.5	29.5	28.0	26.2	20.1	13.8	7.2	(23)
(24)			MCWB	3.8	4.3	8.3	11.3	16.2	19.5	20.6	20.5	20.1	15.6	10.3	5.3	(24)
(25)		10%	DB	3.3	3.8	8.7	14.2	22.4	25.5	27.9	26.6	23.9	17.5	11.5	5.3	(25)
(26)			MCWB	2.0	1.7	4.9	8.5	15.6	18.0	19.6	19.7	19.6	14.0	8.6	3.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.8	11.9	13.4	16.3	21.0	23.1	25.2	24.7	24.0	21.3	15.2	11.4	(27)
(28)			MCD B	13.5	13.2	16.6	21.9	26.5	29.0	29.6	28.6	28.8	24.3	17.1	13.2	(28)
(29)		2%	WB	7.8	7.7	10.8	14.6	19.4	21.7	23.6	23.1	22.7	18.5	12.4	8.5	(29)
(30)			MCD B	9.1	10.3	13.0	19.4	25.9	27.3	28.3	27.1	26.5	21.3	15.1	10.0	(30)
(31)		5%	WB	3.9	4.3	7.9	11.9	18.1	20.7	22.5	22.1	21.4	16.2	10.4	5.8	(31)
(32)			MCD B	5.0	6.4	11.7	16.5	23.2	25.4	26.9	25.7	24.8	18.8	13.3	7.2	(32)
(33)		10%	WB	2.1	1.9	5.3	9.1	16.3	19.5	21.5	21.2	20.1	14.7	8.6	3.9	(33)
(34)			MCD B	3.3	3.9	8.5	13.4	20.7	23.9	25.7	24.8	22.8	16.8	11.3	5.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.8	7.2	7.5	8.5	9.3	8.8	9.3	8.9	8.7	7.9	6.7	5.6	(35)
(36)			MCDBR	9.0	10.7	12.0	14.5	13.5	11.6	11.2	10.4	11.2	11.4	10.5	8.3	(36)
(37)			MCWBR	8.4	9.2	8.3	9.4	6.6	5.5	5.0	4.5	5.4	6.9	7.4	7.4	(37)
(38)		5% WB	MCDBR	8.4	10.5	10.6	13.8	11.6	10.7	9.6	8.8	9.9	9.9	9.7	8.3	(38)
(39)			MCWBR	8.1	9.4	8.4	10.3	6.9	6.0	5.1	4.6	5.6	6.9	7.6	7.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.294	0.337	0.337	0.383	0.405	0.442	0.434	0.419	0.386	0.364	0.312	0.296	(40)	
(41)		taud	2.392	2.235	2.387	2.295	2.284	2.204	2.250	2.280	2.355	2.397	2.510	2.449	(41)	
(42)		Ebn at Noon	840	846	904	882	869	834	836	837	840	811	818	805	(42)	
(43)		Edh at Noon	77	107	105	125	130	141	134	126	108	91	69	68	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.41	2.17	3.41	4.44	5.41	5.93	5.92	5.21	4.16	2.51	1.56	1.07	(44)	
(45)		RadStd	0.14	0.23	0.37	0.48	0.48	0.55	0.41	0.42	0.27	0.39	0.23	0.21	0.12	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page